

# HASBY FAHRUDIN

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## SKILLS

- AI Agent | Machine Learning | Computer Vision | Cloud Architecture | Backend Development | Project Management
- Python | C++ | C# | JavaScript | HTML | CSS | Docker | Git | FastAPI | Django | .NET | JIRA | Github Actions
- TensorFlow | PyTorch | Scikit-learn | Transformers | LangChain | spaCy | Milvus | Pandas | OpenCV | AWS

## EXPERIENCE

### Senior AI Engineer Crosscert, Inc

Mar 2025 - Present  
Seoul, South Korea

- Led the development of a Proof-of-Concept Enterprise Document Retrieval solution utilizing multi-agent orchestration, integrating access-controlled document security for enterprise use cases.
- Delivered an intermediary API to automate domain ownership validation for SSL certificate issuance reducing manual verification overhead.
- Delivered an intermediary platform to support SSL registration agents in issuing certificates through partner CA.

### Lead AI Engineer AIBrain, Inc

May 2023 - Mar 2025  
Seoul, South Korea

- Delivered an enterprise-grade agentic SSL technical support agent, serving as both tech lead and project manager.
- Implement an agent orchestration module enabling fallback handling, personalization, and action management which improve robustness of agent behavior.
- Boosted AI agent resolution rate by 2.5× through RAG retrieval optimization via knowledge restructurization.
- Led development of SAIVA, an integrated web and VR platform for football analysis, delivering the MVP with data-driven features enabling pilots with prospected Bundesliga team.
- Architected and prototyped football pass analysis using XGBoost and game-theoretic modeling.
- Scaled video analysis pipeline by adopting containerization and queue-based processing, expanding capacity to virtually unlimited users.
- Ranked 7th in SoccerNet 2023 Jersey Tracking Challenge using a heuristic-based aggregation pipeline built in a week.

### AI Engineer Crosscert, Inc

May 2021 - May 2023  
Seoul, South Korea

- Built a PoC multi-agent imitation learning model to recommend optimal defensive positioning in football matches using spatial-temporal data.
- Implemented a fast artifact detection system for football broadcast videos using feature-based pattern recognition, matching baseline accuracy with lower compute and data needs.
- Developed a PoC GAN-based system to transform NFT assets into diverse artistic styles and extrapolated 3D representations, enhancing visual richness and interaction potential.
- Supervised and mentored 2-3 interns on AI-driven projects, providing guidance and ensuring effective project progress.

### Master's Research Assistant Wireless Network Lab, SeoulTech

Mar 2020 - Sep 2022  
Seoul, South Korea

- Achieved ~10% accuracy improvement in MAML for few-shot head movement prediction in VR tasks, optimizing computation time with Meta-Curvature and Layer-Specific Learning Rates (LSLR).
- Improved real-time VR head movement prediction accuracy by ~12% using an ensemble learning approach.

### Bachelor's Research Assistant IDEMIA

Sep 2018 - May 2019  
Surabaya, Indonesia

- Developed a lightweight facial emotion classifier using hybrid feature extraction, achieving ~70% accuracy with real-time prediction on edge devices.
- Engineered an assistive device for visually impaired users to perceive facial expressions, achieving 79% accuracy in real-world testing.

### Software Engineer IDEMIA

Jun 2019 - Feb 2020  
Jakarta, Indonesia

- Enhanced system reliability for Remote OS Library Loader leading to 30% decrease in production issues.
- Ensure 100% delivery time for software testing task.

## HIGHLIGHTED PROJECTS

**Seoul National University of Science and Technology**

**M. Sc., Electrical and Information Engineering**

- Thesis: Combining Improvements in Model-Agnostic Meta-Learning.
- GPA: 4.25 of 4.50.

**Mar 2020 - Sep 2022**

**Seoul, South Korea**

**Institut Teknologi Sepuluh November**

**B. Eng., Electrical Engineering**

- Thesis: Deep learning based facial expressions recognition system for assisting visually impaired persons.
- GPA: 3.3 of 4.0

**Aug 2015 - Jul 2019**

**Surabaya, Indonesia**

## HIGHLIGHTED PROJECTS

**AI-Powered Tech Support Agents**

[\[link\]](#)

- An enterprise solution powered by LLM-RAG to deploy tech support AI agent.

**SAIVA: Sport AI Virtual Assistant**

[\[link\]](#)

- Designed to give teams and athletes a competitive edge, SAIVA is the brainchild of a collaboration between AI and VR development experts, professional football coaches, and sports scientists.

**Head Motion Prediction in Online VR**

[\[link\]](#)

- Enhanced an online VR environment using Meta-Learning and ensemble classification for quick, accurate head motion prediction with minimal calibration.

**Facex: Lightweight, High-Performance Facial Expression Classifier**

[\[link\]](#)

- Python library for detecting faces and classifying emotions in images lightweight, efficient threading and object pooling for concurrent processing making it suitable for high-performance application.

## PUBLICATIONS

**SoccerNet 2023 Challenges Results**

[\[link\]](#)

**CVPR 2023**

- Contributing on proposing pipe-line to extract jersey number of each player on football match broadcast video.

**Deep Learning Based Facial Expressions Recognition System for Assisting Visually Impaired persons**

[\[link\]](#)

**Bulletin of Electrical Engineering and Informatics 2020**

- Developed a deep learning-based facial expression recognition system on a wearable device to help visually impaired individuals interpret facial expressions during communication

## OTHERS

- Peer-reviewed papers for **CVSport** 2024 and **ICANN** 2024.
- Represented SAIVA at **Sports Innovation** 2024 (Düsseldorf, Germany) and **VIVATECH** 2024 (Paris, France).
- **Speaker** Regional Scrum Gathering<sup>SM</sup> Seoul 2024, Enhancing Product Management with ChatGPT